

Question

Compound **A** can be prepared by reacting 2,4-disulfophenol with concentrated nitric acid. In 1873, Hermann Sprengel discovered that **A** could be used as a primary explosive, and it was widely employed in subsequent conflicts including the First Sino-Japanese War, the Russo-Japanese War, and World War I. In 1848, Scottish chemist John Stenhouse mixed calcium hypochlorite with a small amount of **A** to produce **B**, which was used by the German military as a tear gas agent during World War I and is now utilized as an insecticide. **B** contains no *H* atoms or *C* – *C* bonds, with $\omega_C = 7.31\%$. Reducing **B** with *Sn* in hydrochloric acid yields another blister agent **C**, which was prepared but never deployed by the German military; its relative molecular mass is less than 150. **C** can also be obtained by reacting the active component of detonators, **D**, with chlorine gas in hydrochloric acid. All atoms in **C** are coplanar, and no charge separation exists.

Electrolyzing a methanol-sulfuric acid solution of **B** yields compound **E** at the cathode. **E** can also be synthesized by reacting nitrosyl chloride with sodium trichloromethanesulfinate. When heated near its melting point, **E** decomposes into equimolar amounts of **B**, **F**, and nitrosyl chloride. Heating at 120°C produces a common solvent **G**, compound **H**, and gas **I** in a molar ratio of 1 : 1 : 2. **F** and **H** have similar structures, with a molecular weight difference of 16.

Deduce the chemical formulas of **A** to **I**. Among the following statements, the incorrect ones are:

1. In the reaction equation for the formation of **B** from **A** and calcium hypochlorite, there are two products containing *Ca*.
2. The conversion of 1 equivalent of **B** to **C** requires the consumption of 2 equivalents of *Sn*.
3. 1 equivalent of **D** produces 2 equivalents of **C**.
4. The formation of 1 equivalent of **E** from **B** consumes 2 equivalents of electrons.
5. No charge separation exists in **H**.
6. The structure of **F** can be regarded as **C** with one substituent replaced.

A. 1

B. 2

C. 3

D. 4

E. 5

F. 6

G. 2,6

H. 4,5

I. 1,2

J. 1,5

K. 3,5

L. 1,5,6

M. 2,4,5

N. 2,3,6

O. None of the above options are correct

Answer

Correct Answer: **J**

Detailed Explanation

Clearly, **A** is trinitrophenol ($C_6H_2(OH)(NO_3)_3$). Since **B** contains no H atoms or $C - C$ bonds, **B** likely contains only one C atom. The relative molecular mass of **B** is $12.01/0.0731 = 164.3$. After subtracting the mass of C , the remaining mass corresponds to 3 Cl atoms and 1 NO_2 group, so **B** = CCl_3NO_2 .

CHECKPOINT

1 PTS

A is trinitrophenol ($C_6H_2(OH)(NO_3)_3$)

CHECKPOINT

1 PTS

B = CCl_3NO_2

The balanced equation for the reaction forming **B** from **A** is as follows:



CHECKPOINT

1 PTS



There are 3 *Ca*-containing products, so statement 1 is incorrect.

C can be obtained by reducing **B** with *Sn* in hydrochloric acid. The fully reduced product of **B** might be CCl_3NH_2 , but this is not a planar compound. Therefore, **C** is likely a partially reduced product of **B**. Since all atoms in **C** are coplanar, a $C = N$ double bond is probable, suggesting **C** could be $Cl_2CN(OH)$ or Cl_2CNH , which is consistent with the given information.

CHECKPOINT

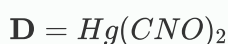
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The active substance in detonators, **D**, could be hydrazoic acid or mercury fulminate. Based on the structure of **C**, the latter is more plausible, i.e., $\mathbf{D} = Hg(CNO)_2$.

CHECKPOINT

1 PTS



When **D** reacts with Cl_2 in hydrochloric acid to form **C**, if **C** were Cl_2CNH , it would be difficult to account for the oxygen atoms from mercury fulminate. Therefore, **C** is $\mathbf{C} = Cl_2CN(OH)$.

CHECKPOINT

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Excluding Cl_2CNH , **C** is $\mathbf{C} = Cl_2CN(OH)$.

The equation for the reduction of **B** to **C** with *Sn* in hydrochloric acid is as follows:



CHECKPOINT

1 PTS



1 equivalent of **B** requires 2 equivalents of *Sn* to form **C**, so statement 2 is correct.

The equation for the reaction of **D** with Cl_2 in hydrochloric acid to form **C** is as follows:



CHECKPOINT

1 PTS

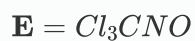


1 equivalent of **D** produces 2 equivalents of **C**, so statement 3 is correct.

Sodium trichloromethanesulfinate can serve as a trichloromethylation reagent. Since **E** can be prepared from nitrosyl chloride and sodium trichloromethanesulfinate, the chemical formula of **E** is Cl_3CNO .

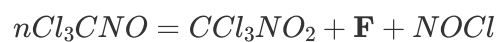
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1 PTS



Reducing 1 equivalent of **B** to **E** consumes 2 equivalents of electrons, so statement 4 is correct.

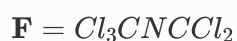
The equation for the thermal decomposition of **E** near its melting point is:



From the number of *O* atoms in the products, $n \geq 3$. When $n = 3$, $\mathbf{F} = \text{C}_2\text{Cl}_5\text{N}$. Based on the structure of **E**, the structure of **F** is $\text{Cl}_3\text{CNCCl}_2$.

CHECKPOINT

1 PTS



The structure of **F** resembles that of **C**, with the hydroxyl group replaced by a trichloromethyl group, so statement 6 is correct.

H has a relative molecular mass 16 units higher than **F**, so $\mathbf{H} = \text{Cl}_3\text{CN}(\text{O})\text{CCl}_2$.

CHECKPOINT

0.5 PTS



The reaction equation for **E** at 120°C is $\text{Cl}_3\text{CNO} = \text{CCl}_4 + \text{Cl}_3\text{CN}(\text{O})\text{CCl}_2 + \text{NO}$, with a product ratio of 1 : 1 : 2.

The $\text{N} - \text{O}$ bond in **H** exhibits charge separation, so statement 5 is incorrect.

CHECKPOINT

0.5 PTS

Charge separation exists in **H**, so statement 5 is incorrect

In summary, statements 1 and 5 are incorrect, so the answer is J.